

action had to happen locally. This is where edge computing assumes significant importance. Suppose you want to turn on the air-conditioner at 9 o'clock, but the power or the network is down at that time and resumes at 9:30. The system should have the intelligence that there is a missed schedule at 9:00 and now it is 9:30. So the AC should turn on now. All this computation has to happen on the edge. So the edge should be very smart."

Zenatix started off with an open source stack for WattMan and enhanced it to suit the applications as per their requirements. This gives WattMan the scalability for customisation and allows the team to deploy the stack on any cloud server—be it a platform-as-a-service like Amazon Web Services or any personalised server. The AI suite uses machine learning algorithms to provide insights using data procured from the systems and raises alerts in case any appliance needs maintenance.

While WattMan provides predictive feedback on possible appliance malfunctions, the Zenatix team also ensures that Wattman's own hardware is working properly and can be managed from a remote location. The hardware is built to accommodate a VPN connection, which gives the administrator access to every end point throughout all the customer sites from a central office. WattMan consistently updates the team on its health by providing data on numerous functional parameters.

"We collect a lot of information on the health of our hardware over the endpoints, like Wi-Fi connectivity status, number of reboots and memory status to name a few. Over time, this helps us to troubleshoot without losing any time and avoid hardware failure," says Amarjeet.

Market presence

WattMan has been deployed at more than 800 sites across the country, setting up roughly 20,000 end

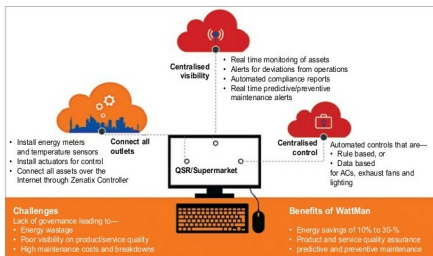


Fig. 2: How WattMan manages retail stores

points. It targets a large range of customers including cold storages, retail chains, bank branches, ATMs and quick service restaurants.

Other than Mother Dairy, Dominos is also using WattMan solution. Zenatix claims that WattMan addresses three major problems faced by Dominos: energy wastage, poor store visibility and high maintenance costs. WattMan is performing energy management and predictive maintenance in 200 outlets of Mother Dairy. Zenatix claims WattMan yields 10-30 per cent cost savings for its customers, including energy savings and reduction in maintenance costs.

The challenges

The Zenatix team recollects the biggest challenge was to customise the hardware to environmental conditions of India and other emerging economies.

"Specifically, the environment in such economies is unfavourable with voltage fluctuations, intermittent network connectivity, power failure, higher noise in the environment and so on. These challenges are non-standard in developed economies like the US or Europe. Therefore importing the hardware from these regions and expecting it to work in our environment was not an option," Amarjeet shared.

The scope of services

Zenatix plans to take WattMan beyond the boundary of energy efficiency.

Amarjeet elucidates, "Today, we are providing energy management solutions to our customers like Dominos. Tomorrow, we might enable them to integrate and control their menus centrally and digitally with the help of WattMan. In fact, we have received similar requests from customers and we are yet to plan our roadmap. Wattman is also benefiting ATMs by improving their uptime with the processor-driven edge system and intelligent design. The key is to design the system such that we can scale up the application as per requirement on the existing hardware. So we have over-invested in the hardware."

With plans to increase the application domain and scalability of WattMan, Zenatix aims to push its business offshore as well. After attending the Microsoft Accelerator programme in 2017, it has partnered with them to leverage their relationships in retail and banking.

Amarjeet mentions, "To this effect, we were recently selected by IoT Acceleration Consortium (ITAC) as one of the ten startups from India to exhibit at CEATEC to meet Japanese companies and government officials and work out synergies." **EFY**